Supplementary Material for "A Structural Contract for Audit Records in Budget-Aware Knowledge-Graph Maintenance"

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This supplement reports synthetic rule-conformance sensitivity (Appendix A), fair-v1 policy-consumption statistics (Appendix B), and the record schema plus offline reproduction contract (Appendix C). Process-annotation calibration and external representation validity fall outside its scope. All reported conformance scores use targets generated by the declared structural rules and therefore provide no independent ground truth.

1 Appendix A. Synthetic scalability and threshold sensitivity

1.1 Frozen construction

The Countries-KG label cache and synthetic graphs use seed 20260711. The synthetic family consists of planted directed dependency graphs with redundant twins. It checks whether the implemented rules remain internally consistent as graph size increases. These graphs do not represent independent production KGs. Table 1 reports the frozen readout.

Table 1 Synthetic rule-conformance and mean path length. The 30-node row is the Countries-KG anchor; all other rows are seeded synthetic DAGs.

Nodes	Bottleneck F1	Redundancy F1	Mean covered path length
30	1.000	1.000	1.042
100	1.000	1.000	2.587
200	1.000	1.000	2.819
500	1.000	1.000	3.115
1000	1.000	1.000	3.349
5000	1.000	1.000	3.883

The perfect F1 values follow from comparing the extractor with labels generated by the same declared structural rules. They are implementation checks. The increasing path length records deeper generated dependency chains and does not measure external performance.

1.2 Redundancy threshold

The declared redundancy edge rule is strict Jaccard similarity greater than 0.85. Repeating the controlled check at 0.90 leaves Countries-KG redundancy support at 124 and macro F1 at 1.000. The synthetic sizes from 100 to 5000 nodes also retain macro F1 1.000 under both thresholds. This invariance reflects the planted fixture margins and does not show that arbitrary semantic KGs are insensitive to the threshold.

2 Appendix B. Policy-consumption statistics, Pareto analysis, and utility

2.1 Fair-v1 protocol and statistical units

All methods share a fixed candidate set and budget within each paired unit. Countries-KG uses the rule `auditable == true`; Wikidata uses `layer > 0 and downstream_impact_count > 0`. Every report stores the sorted candidate identifiers and their SHA-256 digest. The Countries-KG report contains 600 executions, but inference first averages repetitions within each of 30 source traces; the statistical unit is therefore the source trace. Wikidata uses 30 motif seeds on one fixed substrate, so its uncertainty is conditional on that substrate. The 16 anchor clusters are likewise conditional units, not independent KGs.

For paired comparisons, let $d_i = C_i^{\mathrm{LSF}} - C_i^{\mathrm{baseline}}$. We report the paired mean difference, a 10,000-resample percentile interval obtained by resampling unit indices, a two-sided Wilcoxon signed-rank test, and matched-pairs rank-biserial correlation. The main-budget Holm family contains exactly LSF versus Flat Top-K, Greedy Maximum Coverage, and Degree Centrality.

Table 2 Countries-KG fair-v1 policy consumption at $K = 25\%$. Intervals are paired-bootstrap percentile intervals over 30 source traces.

Method	Impact Coverage	95% CI	Mean path length
Life-Saving First	0.733	[0.567, 0.867]	0.828
Greedy Maximum Coverage	0.733	[0.567, 0.867]	0.828
Flat Top-K	0.733	[0.567, 0.867]	0.828
Degree Centrality	0.650	[0.494, 0.794]	0.761
Random Stratified	0.733	[0.567, 0.867]	0.828
Position	0.000	[0.000, 0.000]	0.000
Random	0.364	[0.279, 0.443]	0.593

2.2 Countries-KG policy readout

Life-Saving First, Greedy Maximum Coverage, and Flat Top-K tie exactly on mean Impact Coverage. Random Stratified is non-informative because the eligible structural labels are invariant under within-source shuffling. The no-fallback policy changes selected identifiers in 60 of 600 executions but leaves mean coverage unchanged and uses only 95% of the nominal budget on average. Critical Bottleneck and Unique Evidence have zero activation across all 600 executions; they are unexercised, so Countries-KG does not validate the complete four-layer policy. Redundancy Group Samples is exercised in all 600 executions and Fallback in 60.

2.3 Wikidata main budget and layer ablations

Table 3 Wikidata fair-v1 primary budget over 30 paired motif seeds. Values are mean ± SD.

Method	C	P	Sink-drop mass
Life-Saving First	0.5844 ± 0.0018	0.7859 ± 0.0036	0.7313 ± 0.0033
LSF-Clustered	0.5855 ± 0.0016	0.7873 ± 0.0031	0.7326 ± 0.0029
Greedy Maximum Coverage	0.6329 ± 0.0012	0.8212 ± 0.0029	0.7595 ± 0.0027
Flat Top-K	0.4949 ± 0.0024	0.5867 ± 0.0040	0.5426 ± 0.0037
Degree Centrality	0.4727 ± 0.0030	0.5850 ± 0.0051	0.5410 ± 0.0047

Greedy Maximum Coverage is higher than LSF on both C and P. It is therefore the only nondominated point among the four prespecified main-budget methods. This is a negative policy result for the illustrative LSF consumer: the fair-v1 evidence does not support an LSF governance trade-off or overall advantage.

Removing the bottleneck layer lowers C to 0.5313 and P to 0.6540 at the primary budget. Removing the redundancy layer or the unique-evidence layer does not change the primary-budget selection or metrics. Across all 30 seeds and eight budgets, the bottleneck layer selects 55,376 records and the unique-evidence layer 2,924;

Table 4 Prespecified main-budget paired comparisons, LSF minus baseline. Holm adjustment is applied only within these three rows.

Baseline	Mean diff.	95% CI	Rank-bis.	Holm p
Flat Top-K	0.0895	[0.0889, 0.0901]	1.000	4.78×10^{-6}
Greedy Coverage	-0.0485	[-0.0489, -0.0481]	-1.000	4.78×10^{-6}
Degree Centrality	0.1117	[0.1109, 0.1125]	1.000	4.78×10^{-6}

the redundancy-group and fallback layers select zero. Those two layers are unexercised on Wikidata, so the experiment does not validate the complete four-layer policy. No-fallback is consequently non-informative across the Wikidata sweep and is omitted from result tables.

2.4 Noise inference and anchor-cluster confirmation

Noise outcomes are evaluated on the clean reference graph. Inference is restricted to 20% deletion/insertion, C and P, and LSF versus Flat/Greedy, forming one family of eight comparisons. Table 5 reports paired mean degradation differences; positive values mean LSF degrades less than the baseline. Holm values are generated from the single eight-comparison family in `noise_inference_family.json`.

Table 5 Predeclared 20% clean-reference noise family with Holm adjustment across all eight rows.

Noise	Metric	Baseline	Mean difference	95% CI	Holm p
Deletion	C	Flat	-0.0219	[-0.0260, -0.0176]	2.01×10^{-5}
Deletion	C	Greedy	-0.0030	[-0.0066, 0.0011]	0.0672
Deletion	P	Flat	-0.0261	[-0.0324, -0.0195]	2.36×10^{-5}
Deletion	P	Greedy	0.0072	[0.0021, 0.0122]	0.0373
Insertion	C	Flat	-0.0404	[-0.0459, -0.0354]	1.39×10^{-5}
Insertion	C	Greedy	-0.0211	[-0.0267, -0.0155]	2.11×10^{-5}
Insertion	P	Flat	-0.0080	[-0.0138, -0.0025]	0.0373
Insertion	P	Greedy	0.0167	[0.0104, 0.0227]	7.29×10^{-4}

At $K = 5\%$, the 16 anchor clusters remain conditional subsets of the same substrate. LSF-Clustered selects the same identifiers as LSF in every cluster, so it is recorded as non-informative and omitted from the result table. Table 6 includes the required Greedy baseline.

Greedy also exceeds LSF in the cluster confirmation (paired mean difference -0.0234, Holm $p = 0.0020$). These cluster rows reinforce the negative policy diagnosis conditional on the substrate; they do not support cross-KG generalization.

Table 6 Conditional anchor-cluster confirmation at $K = 5\%$; the 16 units come from one Wikidata substrate.

Method	C mean $\pm$ SD	P mean	LSF minus method, 95% CI
Life-Saving First	0.5313 ± 0.0763	0.5038	–
Greedy Maximum Coverage	0.5546 ± 0.0602	0.5272	[-0.0326, -0.0143]
Flat Top-K	0.5172 ± 0.0928	0.4892	[0.0051, 0.0246]
Degree Centrality	0.4371 ± 0.1271	0.4199	[0.0654, 0.1242]

2.5 Pareto and λ analysis

For a governance-specific diagnostic view only, define

$$U_\lambda(S_K) = \lambda C(S_K) + (1-\lambda)P(S_K), \qquad \lambda \in [0,1]. \tag{1}$$

This scalarization is not part of the SCAR schema and is not used to tune any reported method. The observed main-budget points are $(C, P) = (0.5844, 0.7859)$ for LSF, $(0.6329, 0.8212)$ for Greedy Coverage, $(0.4949, 0.5867)$ for Flat Top-K, and $(0.4727, 0.5850)$ for Degree Centrality. Greedy Coverage strictly dominates LSF in both coordinates, so there is no crossover on the $\lambda = 0, 0.01, \ldots, 1$ grid. Across 10,000 paired bootstrap resamples, the valid crossover proportion is 0 for every prespecified baseline; the 95% reporting rule is not met, and no stable λ^* is reported.

A separate Greedy U_λ diagnostic uses $\lambda = 0, 0.05, \ldots, 1$ and privileged access to both declared evaluation fields during set construction. It serves only as an oracle/Pareto reference. It is neither a baseline for the main claim nor evidence of an implementable universal consumer.

3 Appendix C. Record schema, cache lock, and offline reproduction

3.1 SCAR 1.0 interface

The machine-readable contract is `schemas/scar_audit_record.schema.json`. It uses JSON Schema Draft 2020-12, fixes `schema_version` to `scar-1.0`, rejects unknown properties, and requires the fields in Table 7.

The redundancy graph places an undirected edge between records when Jaccard similarity of terminal coverage is strictly greater than 0.85. Union–find then serializes each connected component as one `redundancy_group_id`. This is a transitive closure: two records can share a group through an intermediate record even when their direct pairwise similarity does not exceed 0.85.

Countries-KG and Wikidata both emit `audit_records.jsonl`; every line is validated against the same schema in the test suite. The Countries records use 30 original source traces, not the 600 execution rows. Wikidata records use the primary frozen audit overlay and include its GraphML SHA-256.

Table 7 Required SCAR 1.0 audit-record fields.

Field	Contract
artifact_id	Stable artifact identifier within the frozen source unit.
graph_snapshot	Snapshot identifier and SHA-256 digest.
auditable	Whether the artifact belongs to the candidate universe.
is_bottleneck	Declared structural bottleneck flag.
is_redundant, redundancy_group_id	Redundancy flag and nullable union-find component identifier.
downstream_impact_count	Number of reachable descendants.
sink_drop_count, at_risk_terminal_ids	Lost terminal count and identifiers under node removal.
raw_risk_score	Trace-local normalized impact field in $[0, 1]$.
extractor_metadata	Extractor, protocol, candidate rule/hash, source unit, and replicate identifiers.

3.2 Frozen inputs and commands

The fair-v1 configuration is `configs/jiis_controlled_maintenance_fair_v1.yaml`. It locks the original 30 motif seeds, budgets from 5% to 50%, deletion/insertion rates from 0 to 20%, 10,000 paired bootstrap rounds, the candidate rule, and the corrected multidisciplinary v2 cache SHA-256. The cache is loaded with `offline: true`; if the file is missing or its digest differs, execution fails before any network request. Revision-history retrieval is skipped in offline mode.

The repository commands are:

```
python scripts/run_countries_kg_label_validation.py
python scripts/run_jiis_audit_case.py --n-traces 600 \
  --seed 20260711 --budget 0.25
python scripts/run_wikidata_scientist_audit.py \
  --config configs/jiis_controlled_maintenance_fair_v1.yaml
python scripts/build_jiis_submission_package.py
python scripts/verify_jiis_submission.py --json \
  paper/JIIS_submission/reports/jiis_verification_report.json
```

3.3 Archive scope

`requirements-jiis-lock.txt` pins the exact Python package versions used for the archived run. The deterministic archive builder writes entries with fixed timestamps, sorts all internal paths, and adds `SHA256SUMS.txt`. The archive contains the fair-v1 configuration, frozen Countries-KG and Wikidata inputs, SCAR schema, source code, audit records, seed-level fair results, and offline commands. It does not claim full reproducibility of deleted process-annotation experiments or any external validation result.

The source TeX sets the required figure search path and uses bare image names. The builder copies only referenced PNG files to the flat package root and excludes LaTeX auxiliaries, unreferenced images, old outputs, and subdirectories.